

SAGA, Japan

WMO: 478130

Lat: 33.2664N

Lon: 130.3053E

Elev: 33

StdP: 100.93

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	0.1	-8.4	1.8	5.4	-6.8	2.1	5.3	10.4	7.5	9.3	7.1	2.2	70	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	34.7	25.4	33.6	25.3	32.5	25.0	26.9	31.6	26.4	30.9	26.0	30.2	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.2	28.9	25.3	20.6	28.5	24.8	19.9	28.3	85.0	31.7	82.7	30.9	80.6	30.3	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.3	7.3	-3.2	36.6	1.0	1.1	-3.9	37.4	-4.5	38.0	-5.1	38.7	-5.9	39.4		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-4.1	27.4	0.9	0.8	-4.8	28.0	-5.3	28.5	-5.9	29.0	-6.6	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	5.8	7.2	10.4	15.4	20.4	23.9	27.8	28.8	25.1	19.6	13.5	8.0
	DBStd	8.34	2.57	3.12	3.10	3.04	2.30	2.06	2.23	1.92	2.54	2.82	3.35	3.06
	HDD10.0	339	132	89	32	2	0	0	0	0	0	0	7	78
	HDD18.3	1534	388	312	246	95	8	0	0	0	0	18	147	320
	CDD10.0	2975	2	11	45	164	322	416	553	582	452	299	112	16
	CDD18.3	1127	0	0	0	7	71	166	294	324	203	59	3	0
	CDH23.3	10370	0	0	0	39	435	1076	3190	3768	1591	266	6	0
	CDH26.7	3829	0	0	0	2	80	266	1292	1656	499	35	0	0
Wind	WSAvg	3.1	2.8	3.1	3.3	3.1	3.0	3.1	3.3	3.2	3.2	3.1	2.7	2.8
Precipitation	PrecAvg	1803	65	78	113	141	166	302	308	214	169	86	97	65
	PrecMax	2527	188	143	261	276	341	510	581	574	562	200	189	136
	PrecMin	951	27	29	57	49	43	35	19	41	64	10	15	10
	PrecStd	342	33	33	43	61	78	139	171	128	103	57	48	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	18.9	21.9	26.4	30.7	32.1	36.0	36.2	33.9	29.3	24.2	18.0
		MCWB	11.4	13.6	14.8	16.7	18.2	21.5	26.1	25.9	24.6	21.8	17.5	13.5
	2%	DB	13.3	16.4	19.8	24.4	28.7	30.7	34.5	35.1	32.4	27.5	22.1	16.2
		MCWB	9.6	12.1	13.4	16.0	18.2	21.5	25.4	25.6	24.2	20.3	16.5	11.8
	5%	DB	11.9	14.4	17.9	22.8	27.3	29.5	33.4	34.1	31.0	26.1	20.5	14.7
		MCWB	8.4	10.3	12.6	15.2	17.8	21.6	25.3	25.5	23.6	19.5	15.6	11.1
	10%	DB	10.6	12.7	16.1	21.2	25.9	28.2	32.2	33.0	29.6	24.7	19.1	13.3
		MCWB	7.2	8.9	11.6	15.0	17.6	21.6	25.3	25.4	23.0	18.7	14.9	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	15.4	16.7	20.6	22.7	25.6	27.6	27.7	26.5	24.2	19.8	15.4
		MCDB	14.2	17.0	18.8	22.7	25.2	28.2	32.6	32.3	30.3	27.1	22.1	16.0
	2%	WB	10.8	13.2	15.1	18.8	21.6	24.8	26.9	27.1	25.6	22.7	18.1	13.2
		MCDB	12.2	15.3	17.8	21.4	23.8	27.6	31.7	31.7	29.5	25.8	20.0	15.0
	5%	WB	9.0	11.3	13.7	17.6	20.5	24.2	26.4	26.6	24.9	21.2	16.9	11.6
		MCDB	11.2	13.5	16.9	20.6	23.5	26.9	30.7	31.2	28.9	24.7	19.5	13.8
	10%	WB	7.6	9.5	12.4	16.3	19.5	23.4	25.9	26.1	24.2	19.8	15.6	10.3
		MCDB	10.2	12.1	15.5	19.7	23.4	26.1	30.0	30.6	28.0	23.0	18.6	13.0
Mean Daily Temperature Range	5% DB	MDBR	7.6	8.2	8.9	9.6	9.7	7.4	6.9	7.6	7.8	8.8	8.5	7.8
		MCDBR	9.3	10.4	11.3	12.1	12.4	10.0	9.1	9.2	9.4	10.1	9.7	9.4
		MCWBR	6.3	6.8	6.9	6.0	5.3	3.8	3.1	3.0	3.6	4.5	5.3	6.0
	5% WB	MCDBR	7.8	8.3	8.8	8.3	7.6	5.7	7.6	7.8	7.5	7.3	7.4	7.7
MCWBR		6.4	6.9	6.8	5.5	4.7	2.9	2.9	2.9	3.2	4.1	5.1	6.0	
Clear-Sky Solar Irradiance	taub	0.429	0.470	0.529	0.548	0.557	0.571	0.561	0.533	0.504	0.457	0.428	0.421	
	taud	2.061	1.989	1.845	1.853	1.892	1.942	2.013	2.088	2.124	2.158	2.152	2.096	
	Ebn at Noon	751	763	750	759	756	743	748	763	766	769	750	732	
	Edh at Noon	133	159	197	204	198	188	175	160	148	132	121	122	
All-Sky Solar Radiation	RadAvg	2.20	2.91	3.99	4.80	5.18	4.38	4.90	5.14	4.17	3.53	2.61	2.04	
	RadStd	0.21	0.28	0.27	0.41	0.59	0.59	0.71	0.53	0.43	0.36	0.25	0.14	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page